Ishaan Meena

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Education

University of Alberta — Edmonton, Canada
Bachelor of Science in Computer Science | GPA: 3.60 out of 4 — *Sep. 2022 – Dec 2027*

Experience

Huawei Technologies Canada – Co-op Researcher, Computer Vision and Machine Learning — *May 26 – Present*

- Developed a state-of-the-art feed-forward 3D scene reconstruction system in PyTorch that generates object-level meshes and 3D layouts from posed multi-view images, achieving SOTA performance against adjacent methods.
- Designed a visibility- and occlusion-aware geometry-completion architecture combining Transformers, 3D CNNs, cross-view evidence, and generative 3D priors; reduced Scene CD by 31.5% and Scene CD-S by 36.8%, while improving Scene F-score by 3.67 points over the matched non-occlusion ablation.
- Built reproducible preprocessing, training, benchmarking, and ablation pipelines spanning ground-truth and predicted-mask settings; evaluated recent 3D reconstruction methods and analyzed 156 scenes to isolate failures in segmentation, object layout, and shape completion.
- Preparing an ICLR 2027 paper and accompanying research codebase, translating literature reviews and quantitative failure analysis into novel architectural contributions and presenting findings through weekly and monthly research reviews.

PCL Applied Research AI Capstone Project — *Jan – May 26*

- Developed an LLM-powered natural-language interface for BIM digital twins, combining Text-to-SQL and Graph-RAG-based information retrieval for multi-hop spatial reasoning and entity-centric question answering. Implemented hybrid retrieval using Text-to-SQL and Graph-RAG for multi-hop spatial reasoning.
- Produced research-style evaluations analyzing accuracy, failure modes, and graph traversal behavior.

Steadfast Engineering: Software Engineering Intern | *Next.js, React, TypeScript, Supabase* — *Sept – Dec 25*

- Developed a full-stack project management system for Steadfast Engineering using **Next.js, React 19, TypeScript, Drizzle ORM, Supabase, and Vercel**, streamlining workflow for **250+ annual projects** and reducing manual tracking errors by **~40%**.
- Implemented **secure authentication, role-based access control, time tracking, payroll exports (Excel/PDF), budget/KPI dashboards**, and real-time team workload monitoring, enabling **leads and admins to manage 50+ concurrent projects** efficiently.
- Designed and deployed **data migration pipelines, Gantt-based project scheduling, and KPI reporting APIs**, maintaining historical records from **2016 onward**, improving decision-making speed and accuracy by **~35%** for management.

Projects

Diffusion & Flow Matching (DDPM + FM) on CIFAR-10

- Built and trained a **latent-space DDPM** using a fully connected autoencoder + time-embedded UNet on **50k CIFAR-10 images**, implementing full forward and reverse diffusion; generated samples meeting expected benchmark quality (IS **1.5–2**, FID **300–350**).
- Implemented a **Flow Matching model** with time-embedded UNet and ODE-based reverse sampling.

LexiServe – AI-Powered Text Intelligence API Platform | *FastAPI, React, PostgreSQL, Docker, Transformers*

- Developed a full-stack **NLP-as-a-Service** system for grammar correction, sentiment analysis, and summarization.
- Integrated fine-tuned **BERT, T5, and DistilGPT2** models via TorchScript for <**100 ms inference latency**.
- Built a React dashboard for real-time analytics and PostgreSQL-backed persistence.
- Deployed scalable, containerized microservices (FastAPI + Celery + Redis) on AWS EC2, handling **10K+ API calls/day**.

YOLO Object Detection – High-Performance Deployment | *PyTorch, YOLOv5, CUDA, FastAPI, Docker*

- Adapted **YOLOv5** for multi-object detection in high-resolution RGB images under varied noise and lighting.
- Achieved **95% mAP at 30 FPS with GPU** acceleration, benchmarking via **IoU, precision/recall**, and confusion matrices.
- Optimized inference using **CUDA kernels**, asynchronous data loading, and **batch preprocessing**, cutting **latency by 45%.**

Resource Locator – Automated Data Ingestion & ETL Pipeline | Python, Amazon Aurora, Web Scraping

- Developed an automated **data ingestion and indexing pipeline** storing 100K+ records with <200 ms query latency.
- Built scalable **ETL processes** for structured resource retrieval and transformation.
- Optimized **data integrity** using transactional batch updates and Aurora indexing, reducing cloud query costs by 25%.

Technical Skills

Programming Languages: Python, C, C++, Java, Kotlin, SQL, JavaScript, TypeScript, HTML/CSS, R
Machine Learning & Data Science: PyTorch, TensorFlow, Scikit-Learn, Transformers, NumPy, Pandas, Matplotlib, OpenAI Gym, MLflow, Kubeflow, CUDA
Backend & API Development: FastAPI, Django REST, Flask, Spring Boot, Go (Gin), Node.js, Express.js, REST APIs, OAuth 2.0
Frontend & Mobile Frameworks: React.js, Next.js, Angular, Vue.js, React Native, Svelte, Tailwind CSS
Databases & Data Systems: PostgreSQL, MySQL, MongoDB, Firebase, Amazon Aurora, Supabase, Redis, AWS S3
Cloud & DevOps: AWS (EC2, Lambda, CloudWatch, S3), Google Cloud Platform, Docker, Vercel, GitHub Actions, Jenkins, GitLab CI/CD
Tools & Platforms: CUDA, OpenCV, Plotly, Drizzle ORM, Celery, Redis Queue, Streamlit, WebRTC